

ECON1050 Tutorial 7

Partial Derivatives & Multivariate Optimisation

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Today's Plan

- ▶ Recap: key ideas from Lecture 7
- ▶ Tutorial questions: Q1–Q4

Recap: Functions of Two Variables

Definition

A function of two variables assigns one real number to each ordered pair (x, y) in its domain:

$$z = f(x, y).$$



Variables, domain and range

- ▶ x and y are independent (exogenous) variables.
- ▶ z is the dependent (endogenous) variable.
- ▶ The domain is a set of ordered pairs (x, y) in the x - y plane.
- ▶ The range is the set of corresponding values of z .

Recap: First-Order Partial Derivatives

Partial derivatives

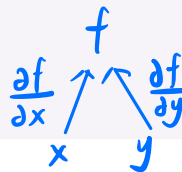
For $z = f(x, y)$,

$$f_x, \frac{\partial f}{\partial x}$$

$$f_x(x, y) = \frac{\partial f}{\partial x}$$

and

$$f_y(x, y) = \frac{\partial f}{\partial y}$$



Interpretation

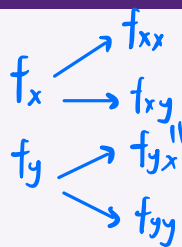
- ▶ f_x measures the rate of change in f with respect to x , holding y constant.
- ▶ f_y measures the rate of change in f with respect to y , holding x constant.

Recap: Second-Order Partial Derivatives

Four second-order partial derivatives

For $f(x, y)$,

$$\begin{aligned}f_{xx} &= \frac{\partial^2 f}{\partial x^2}, & f_{yy} &= \frac{\partial^2 f}{\partial y^2}, \\f_{xy} &= \frac{\partial^2 f}{\partial y \partial x}, & f_{yx} &= \frac{\partial^2 f}{\partial x \partial y}.\end{aligned}$$



The last two are called mixed partial derivatives.

Mixed partials

Under the regularity conditions used in this course,

$$\boxed{f_{xy} = f_{yx}}.$$

Recap: Economic Interpretation of Cross-Price Effects

Cross-price effects

If demand for good A depends on the price of good B , then

$$\frac{\partial q_A}{\partial p_B}$$

measures how demand for A responds to the price of B .

Substitutes vs complements

Suppose demand for good A depends on the price of good B .

$$\frac{\partial q_A}{\partial p_B} > 0 \implies A \text{ and } B \text{ are substitutes, } \text{competitive}$$

$$\frac{\partial q_A}{\partial p_B} < 0 \implies A \text{ and } B \text{ are complements.}$$

Recap: Multivariate Optimisation and FOCs

Interior stationary point

For a differentiable function $z = f(x, y)$, an interior maximum or minimum can occur only at a stationary point satisfying

$$\boxed{f_x(x_0, y_0) = 0, \quad f_y(x_0, y_0) = 0}. \Rightarrow \text{stationary points}$$

These are the first-order conditions (FOCs).

Important

The FOCs identify candidates for interior extrema, but they do not by themselves classify the stationary points.

Recap: Second-Order Conditions for Local Extrema

At an interior stationary point (x_0, y_0) , define

$$A = f_{xx}(x_0, y_0), \quad B = f_{xy}(x_0, y_0), \quad C = f_{yy}(x_0, y_0),$$

and

f_{xx}

f_{xy}

f_{yy}

$$D = AC - B^2.$$

$f_{xx}f_{yy} - f_{xy}^2$

Classification

$A < 0, D > 0 \implies$ strict local maximum,

$A > 0, D > 0 \implies$ strict local minimum,

$D < 0 \implies$ saddle point,

$D = 0 \implies$ test is inconclusive.

Recap: Concavity, Convexity and Global Extrema

Convex set

A set S in the x - y plane is convex if, for any two points in S , the full line segment between them also lies in S .

Sufficient conditions on a convex set

For a C^2 function $f(x, y)$ defined on a convex set S :

- ▶ If $f_{xx} \leq 0$, $f_{yy} \leq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is concave.
- ▶ If $f_{xx} \geq 0$, $f_{yy} \geq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is convex.

A stationary point of a concave function is a global maximum; a stationary point of a convex function is a global minimum.

Recap: Extreme Value Theorem in Two Variables

Extreme Value Theorem

If $f(x, y)$ is continuous on a nonempty, closed and bounded set S , then f attains both a global maximum and a global minimum on S .

Finding global extrema on S

1. Find all stationary points in the interior of S .
2. Find the largest and smallest values of f on the boundary of S .
3. Compare all candidate values to identify the global maximum and minimum.

Tutorial Q1

Question 1

Find all first- and second-order partial derivatives for:

1. $z = x^3 y^2$,

② $z = x^5 - 3x^2 y + y^6$,

③ $z = \frac{x}{y}$.

3. $z = \frac{x}{y} = f(x, y)$

$$\begin{aligned} f_x &= \frac{1}{y} \rightarrow f_{xx} = 0 \\ &\rightarrow f_{xy} = -\frac{1}{y^2} \\ f_y &= -\frac{x}{y^2} \rightarrow f_{yx} = -\frac{1}{y^2} \\ &\rightarrow f_{yy} = -x \cdot \frac{-2}{y^3} = \frac{2x}{y^3} \end{aligned}$$

$$\begin{aligned} 2. \quad z &= x^5 - 3x^2 y + y^6 = f(x, y) \\ f_x &= 5x^4 - 3y \cdot 2x + 0 = 5x^4 - 6xy \rightarrow f_{xx} = 5 \cdot 4x^3 - 6y \cdot 1 = 20x^3 - 6y \\ &\rightarrow f_{xy} = 0 - 6x = -6x \\ f_y &= 0 - 3x^2 \cdot 1 + 6y^5 = -3x^2 + 6y^5 \rightarrow f_{yx} = -6x \\ &\rightarrow f_{yy} = 0 + 6 \cdot 5y^4 = 30y^4 \end{aligned}$$

Tutorial Q1(a): $z = x^3y^2$

Knowledge used

- ▶ First-order partial derivatives: differentiate with respect to one variable while holding the other constant.
- ▶ Second-order and mixed partial derivatives.

Solution

First-order partial derivatives:

$$z_x = \frac{\partial z}{\partial x} = 3x^2y^2, \quad z_y = \frac{\partial z}{\partial y} = 2x^3y.$$

Second-order partial derivatives:

$$z_{xx} = 6xy^2, \quad z_{yy} = 2x^3,$$

$$z_{xy} = z_{yx} = 6x^2y.$$

Tutorial Q1(b): $z = x^5 - 3x^2y + y^6$

Knowledge used

- ▶ Partial differentiation term by term.
- ▶ When differentiating with respect to x , treat y as a constant, and vice versa.

Solution

$$z_x = 5x^4 - 6xy, \quad z_y = -3x^2 + 6y^5.$$

$$z_{xx} = 20x^3 - 6y, \quad z_{yy} = 30y^4.$$

For the mixed partials,

$$z_{xy} = z_{yx} = -6x.$$

Tutorial Q1(c): $z = \frac{x}{y}$

Knowledge used

- ▶ Rewrite as $z = xy^{-1}$ and use the power rule.
- ▶ Compute both own-variable and mixed second-order partial derivatives.

Solution

For $y \neq 0$,

$$z_x = \frac{1}{y}, \quad z_y = -\frac{x}{y^2}.$$

Then

$$z_{xx} = 0, \quad z_{yy} = \frac{2x}{y^3}, \quad \boxed{z_{xy} = z_{yx} = -\frac{1}{y^2}}.$$

Tutorial Q2

Question 2

Let $f(x, y) = x^2 + 2xy^2 + 2y^2$.

5 min

1. Find the first- and second-order partial derivatives.
2. Show that the stationary points are $(0, 0)$, $(-1, 1)$ and $(-1, -1)$, and classify them.

Question 2

Let $f(x, y) = x^2 + 2xy^2 + 2y^2$.

1. Find the first- and second-order partial derivatives.
2. Show that the stationary points are $(0, 0)$, $(-1, 1)$ and $(-1, -1)$, and classify them.

At an interior stationary point (x_0, y_0) , define

$$A = f_{xx}(x_0, y_0), \quad B = f_{xy}(x_0, y_0), \quad C = f_{yy}(x_0, y_0), \quad D = AC - B^2.$$

Classification

$A < 0, D > 0 \implies$ strict local maximum,

$A > 0, D > 0 \implies$ strict local minimum,

$D < 0 \implies$ saddle point,

$D = 0 \implies$ test is inconclusive.

Let $f(x, y) = x^2 + 2xy^2 + 2y^2$.

1. Find the first- and second-order partial derivatives.
2. Show that the stationary points are $(0, 0)$, $(-1, 1)$ and $(-1, -1)$, and classify them.

$$1. \quad f(x, y) = x^2 + 2xy^2 + 2y^2$$

$$f_x = 2x + 2y^2 \quad \begin{cases} f_{xx} = 2 \\ f_{xy} = 4y \end{cases}$$

$$f_y = 4xy + 4y \quad \begin{cases} f_{yx} = 4y \\ f_{yy} = 4x + 4 \end{cases}$$

$$A = f_{xx}(x_0, y_0), \quad B = f_{xy}(x_0, y_0), \quad C = f_{yy}(x_0, y_0), \quad D = AC - B^2.$$

Classification

$$\begin{aligned} A < 0, D > 0 &\Rightarrow \text{strict local maximum,} \\ A > 0, D > 0 &\Rightarrow \text{strict local minimum,} \\ D < 0 &\Rightarrow \text{saddle point,} \\ D = 0 &\Rightarrow \text{test is inconclusive.} \end{aligned}$$

$$2. \quad \text{Stationary points: } f_x = 0, f_y = 0$$

$$\begin{cases} 2x + 2y^2 = 0 \\ 4xy + 4y = 0 \end{cases} \Rightarrow \begin{cases} x + y^2 = 0 \quad (1) \\ y \cdot (x + 1) = 0 \quad (2) \end{cases}$$

$$(2): y(x+1) = 0 \quad \begin{cases} y = 0 \\ x+1 = 0 \Leftrightarrow x = -1 \end{cases}$$

$$y = 0: x + 0 = 0 \Rightarrow \text{Stationary point } (0, 0)$$

$$x = -1: -1 + y^2 = 0 \Rightarrow y^2 = 1 \Rightarrow y = \pm 1$$

$$\Rightarrow \text{Stationary points: } (-1, -1), (-1, 1)$$

The stationary points are $(0, 0)$, $(-1, -1)$, and $(-1, 1)$

$$A = f_{xx} = 2$$

$$D = f_{xx} \cdot f_{yy} - f_{xy}^2 = 2 \cdot (4x + 4) - (4y)^2$$

$$= 8x + 8 - 16y^2$$

plug $(0, 0)$ into A and D :

$$A = 2 > 0$$

$$D = 0 + 8 - 16 \cdot 0 = 8 > 0$$

$\Rightarrow (0, 0)$ is a local maximum

plug $(-1, -1)$ into A and D :

$$A = 2 > 0$$

$$D = 8 \cdot (-1) + 8 - 16 \cdot (-1)^2 = -16 < 0$$

$\Rightarrow (-1, -1)$ is a saddle point

plug $(-1, 1)$ into A and D :

$$A = 2 > 0$$

$$D = 8 \cdot (-1) + 8 - 16 \cdot (1)^2 = -16 < 0$$

$\Rightarrow (-1, 1)$ is a saddle point

Tutorial Q2: Partial Derivatives

Knowledge used

- ▶ First- and second-order partial derivatives.
- ▶ Mixed partial derivatives.

Solution

First-order partial derivatives:

$$f_x = 2x + 2y^2, \quad f_y = 4xy + 4y.$$

Second-order partial derivatives:

$$f_{xx} = 2, \quad f_{yy} = 4x + 4, \quad \boxed{f_{xy} = f_{yx} = 4y}.$$

Tutorial Q2: Stationary Points

Knowledge used

- Multivariate FOCs: solve $f_x = 0$ and $f_y = 0$ simultaneously.

Solution

The FOCs are

$$2x + 2y^2 = 0, \quad 4xy + 4y = 4y(x + 1) = 0.$$

From the second equation,

$$y = 0 \quad \text{or} \quad x = -1.$$

If $y = 0$, then $x = 0$. If $x = -1$, then $y^2 = 1$, so $y = \pm 1$. Therefore,

$$(0, 0), \quad (-1, 1), \quad (-1, -1)$$

are the stationary points.

Tutorial Q2: Classification

Knowledge used

- Second-order conditions with $A = f_{xx}$, $B = f_{xy}$, $C = f_{yy}$ and $D = AC - B^2$.

Solution

Here

$$A = 2, \quad B = 4y, \quad C = 4x + 4.$$

(x_0, y_0)	A	B	C	$D = AC - B^2$	Result
$(0, 0)$	2	0	4	8	Local minimum
$(-1, -1)$	2	-4	0	-16	Saddle point
$(-1, 1)$	2	4	0	-16	Saddle point

Hence

$(0, 0)$ is a local minimum; $(-1, \pm 1)$ are saddle points.

Question 3

A monopolist sells two products, q_1 and q_2 , with demand equations

$$0.1p_1 - 1.2 + 0.2q_1 = 0, \quad 10p_2 - 320 + 40q_2 = 0,$$

and joint cost

$$C = q_1^2 + q_2^2 + 2q_1q_2.$$

Find the price and output for each product that maximise profit.

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Sufficient conditions on a convex set

For a C^2 function $f(x, y)$ defined on a convex set S :

- ▶ If $f_{xx} \leq 0$, $f_{yy} \leq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is concave.
- ▶ If $f_{xx} \geq 0$, $f_{yy} \geq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is convex.

A stationary point of a concave function is a global maximum; a stationary point of a convex function is a global minimum.

A monopolist sells two products, q_1 and q_2 , with demand equations

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and joint cost

$$C = q_1^2 + q_2^2 + 2q_1q_2.$$

Find the price and output for each product that maximise profit.

$$\pi = \text{Revenue} - \text{Cost}$$

$$= P \cdot q - C$$

$$= p_1 \cdot q_1 + p_2 \cdot q_2 - (q_1^2 + q_2^2 + 2q_1q_2)$$

$$0.1p_1 - 1.2 + 0.2q_1 = 0, \quad 10p_2 - 320 + 40q_2 = 0,$$

↓

$$p_1 = 12 - 2q_1$$

↓

$$p_2 = 32 - 4q_2$$

$$\pi(q_1, q_2) = (12 - 2q_1) \cdot q_1 + (32 - 4q_2) \cdot q_2 - (q_1^2 + q_2^2 + 2q_1q_2)$$

$$= 12q_1 + 32q_2 - 3q_1^2 - 5q_2^2 - 2q_1q_2$$

For a C^2 function $f(x, y)$ defined on a convex set S :

► If $f_{xx} \leq 0$, $f_{yy} \leq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is concave.

► If $f_{xx} \geq 0$, $f_{yy} \geq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is convex.

A stationary point of a concave function is a global maximum; a stationary point of a convex function is a global minimum.

$$\pi_{q_1} = 12 - 6q_1 - 2q_2 \rightarrow \pi_{q_1} = -6$$

$$\rightarrow \pi_{q_2} = -2$$

$$\pi_{q_2} = 32 - 10q_2 - 2q_1 \rightarrow \pi_{q_2} = -2$$

$$\rightarrow \pi_{q_1q_2} = -10$$

$$\text{Stationary point } \begin{cases} \pi_{q_1} = 12 - 6q_1 - 2q_2 = 0 \\ \pi_{q_2} = 32 - 10q_2 - 2q_1 = 0 \end{cases} \Rightarrow \begin{cases} q_1 = 1 \\ q_2 = 3 \end{cases}$$

$$\Rightarrow \text{Stationary point } (1, 3)$$

$$\pi_{q_1q_1} < 0, \pi_{q_2q_2} < 0, \pi_{q_1q_1} \cdot \pi_{q_2q_2} - \pi_{q_1q_2}^2 = -6 \cdot (-10) - 4 = 56 > 0$$

$$\Rightarrow \pi(q_1, q_2) \text{ is concave at every point}$$

$$\Rightarrow (1, 3) \text{ is a global maximum}$$

Tutorial Q3: Construct the Profit Function

Knowledge used

- ▶ Inverse demand: express each price as a function of its quantity.
- ▶ Profit: $\pi = TR - TC$ with $TR = p_1 q_1 + p_2 q_2$.

Solution

From the demand equations,

$$p_1 = 12 - 2q_1, \quad p_2 = 32 - 4q_2.$$

Therefore,

$$\begin{aligned}\pi(q_1, q_2) &= (12 - 2q_1)q_1 + (32 - 4q_2)q_2 \\ &\quad - (q_1^2 + q_2^2 + 2q_1 q_2) \\ &= 12q_1 + 32q_2 - 3q_1^2 - 5q_2^2 - 2q_1 q_2.\end{aligned}$$

Tutorial Q3: First-Order Conditions

Knowledge used

- ▶ Multivariate profit maximisation: set both first-order partial derivatives equal to zero.
- ▶ Solve the two FOCs simultaneously.

Solution

$$\frac{\partial \pi}{\partial q_1} = 12 - 6q_1 - 2q_2 = 0,$$

$$\frac{\partial \pi}{\partial q_2} = 32 - 10q_2 - 2q_1 = 0.$$

From the first equation,

$$q_2 = 6 - 3q_1.$$

Substituting into the second equation gives

$$32 - 10(6 - 3q_1) - 2q_1 = 0,$$

so

$$\boxed{q_1^* = 1, \quad q_2^* = 3}.$$

Tutorial Q3: Second-Order Conditions and Prices

Knowledge used

- ▶ Concavity/second-order conditions confirm that the stationary point is a maximum.
- ▶ Substitute optimal quantities into inverse demand to recover prices.

Solution

The second-order partial derivatives are

$$\pi_{11} = -6, \quad \pi_{22} = -10, \quad \pi_{12} = -2.$$

Hence

$$\pi_{11}\pi_{22} - \pi_{12}^2 = (-6)(-10) - (-2)^2 = 56 > 0,$$

and $\pi_{11} < 0$. Thus the profit function is concave and the stationary point is the global maximum.

Prices are

$$p_1^* = 12 - 2(1) = 10, \quad p_2^* = 32 - 4(3) = 20.$$

Therefore,

$$\boxed{q_1^* = 1, \quad p_1^* = 10, \quad q_2^* = 3, \quad p_2^* = 20}.$$

Tutorial Q4

Question 4

The weekly demand equations for butter and margarine are

$$Q_b = 8000 - 0.09P_b^2 + 0.008P_m^2,$$

$$Q_m = 15000 + 0.004P_b^2 - 0.03P_m^2.$$

Determine whether the two products are competitive, complementary, or neither.

Tutorial Q4

Question 4

The weekly demand equations for butter and margarine are

$$Q_b = 8000 - 0.09P_b^2 + 0.008P_m^2,$$

$$Q_m = 15000 + 0.004P_b^2 - 0.03P_m^2.$$

4 mins

Determine whether the two products are competitive, complementary, or neither.

Cross-Price Partial Derivatives

Suppose demand for good A depends on the price of good B .

$$\frac{\partial q_A}{\partial p_B} > 0 \implies A \text{ and } B \text{ are substitutes,}$$

$$\frac{\partial q_A}{\partial p_B} < 0 \implies A \text{ and } B \text{ are complements.}$$

The weekly demand equations for butter and margarine are

$$Q_b = 8000 - 0.09P_b^2 + 0.008P_m^2,$$

$$Q_m = 15000 + 0.004P_b^2 - 0.03P_m^2.$$

Determine whether the two products are competitive, complementary, or neither.

Cross-price effect

$$\frac{\partial Q_b}{\partial P_m} = 0.016 \cdot P_m > 0$$

$\Rightarrow$ the two products are substitutes $\Rightarrow$ competitive

$$\frac{\partial Q_m}{\partial P_b} = 0.008 \cdot P_b > 0$$

Tutorial Q4: Cross-Price Partial Derivatives

Knowledge used

- ▶ Cross-price partial derivatives measure how demand for one product changes when the other product's price changes.
- ▶ Positive cross-price effects imply substitutes; negative cross-price effects imply complements.

Solution

The cross-price partial derivatives are

$$\frac{\partial Q_b}{\partial P_m} = 0.016P_m, \quad \frac{\partial Q_m}{\partial P_b} = 0.008P_b.$$

Since prices are nonnegative, both cross-price effects are nonnegative. Thus, when the price of one good rises, demand for the other rises. Therefore,

butter and margarine are competitive products, i.e. substitutes.

Key Takeaways

- ▶ Partial derivatives measure the effect of changing one variable while holding the others constant.
- ▶ Second-order and mixed partial derivatives are central to classifying multivariate stationary points.
- ▶ Interior candidates satisfy the FOCs $f_x = f_y = 0$.
- ▶ The sign of $D = f_{xx}f_{yy} - f_{xy}^2$, together with f_{xx} , distinguishes local maxima, local minima and saddle points.
- ▶ Concavity/convexity can upgrade a stationary point to a global optimum on a convex domain.
- ▶ Cross-price partial derivatives identify substitute and complementary goods.